

# WFLOW-07: Problem-Triggered Remediation

**Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 7 of 9 |  
**Created:** January 2026 | **Last Updated:** 01/28/2026

## Auto-Remediation with Workflows

Move beyond notifications to automated problem resolution. This notebook covers remediation patterns, safety guardrails, runbook automation, and common remediation scenarios.

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## Prerequisites

| Requirement           | Details                         |
|-----------------------|---------------------------------|
| Dynatrace Environment | SaaS with Platform subscription |

| Requirement      | Details                                 |
|------------------|-----------------------------------------|
| Permissions      | <code>automation:workflows:write</code> |
| Prior Knowledge  | <b>WFLOW-01</b> through <b>WFLOW-06</b> |
| External Systems | kubectl access, cloud API credentials   |

# 1. Auto-Remediation Principles

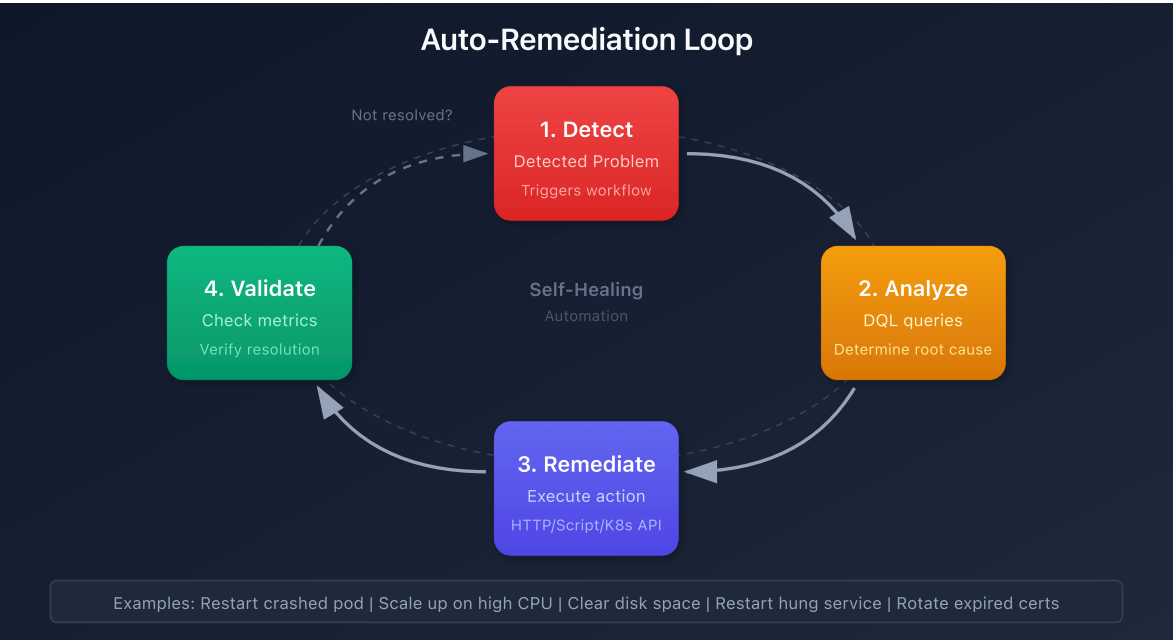
## When to Automate Remediation

| Scenario          | Automate? | Reason                     |
|-------------------|-----------|----------------------------|
| Pod crash loop    | Yes       | Well-defined fix (restart) |
| Disk space 90%    | Yes       | Clear action (cleanup)     |
| Memory pressure   | Maybe     | Scale up vs investigate    |
| Database deadlock | No        | Needs investigation        |
| Security incident | No        | Requires human judgment    |

## Remediation Maturity Model

Level 0: Manual remediation (runbook lookup)  
 Level 1: Automated notification with runbook link  
 Level 2: Semi-automated (human approval required)  
 Level 3: Fully automated with guardrails  
 Level 4: Self-healing with learning

# Auto-Remediation Loop



## 2. Safety Guardrails

### Essential Guardrails

| Guardrail           | Implementation          | Purpose                 |
|---------------------|-------------------------|-------------------------|
| Rate limiting       | Max 3 remediations/hour | Prevent loops           |
| Time window         | Business hours only     | Avoid off-hours changes |
| Environment scope   | Non-prod first          | Protect production      |
| Cooldown period     | 30 min between actions  | Allow stabilization     |
| Rollback capability | Capture pre-state       | Enable undo             |

## Rate Limiting Implementation

```
import { queryExecutionClient } from '@dynatrace-sdk/client-query';

export default async function({ event }) {
  const entityId = event.root_cause_entity_id;

  // Check recent remediation attempts
  const result = await queryExecutionClient.queryExecute({
    body: {
      query: `
        fetch events, from: now() - 1h
        | filter event.type == "automation.workflow.execution"
        | filter contains(workflow.name, "remediation")
        | filter entity_id == "${entityId}"
        | summarize attempts = count()
      `
    }
  });

  const attempts = result.result.records[0]?.attempts || 0;
  const maxAttempts = 3;

  if (attempts >= maxAttempts) {
    return {
      proceed: false,
      reason: `Rate limit reached: ${attempts}/${maxAttempts} in last hour`
    };
  }

  return { proceed: true, attempt_number: attempts + 1 };
}
```

## Time Window Check

```
conditions:
  - name: is_safe_window
    expression: |
      {% set hour = now().hour %}
      {% set weekday = now().weekday() %}
      {{ weekday < 5 and hour >= 6 and hour < 22 }}

tasks:
  - name: remediation_action
    conditions: [is_safe_window]
    # ... action config
```

### 3. Common Remediation Patterns

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#### Pattern 1: Service Restart

```
// Restart a service via SSH/API
export default async function({ event }) {
  const hostId = event.root_cause_entity_id;
  const serviceName = extractServiceName(event.title);

  // Call external automation system
  const response = await fetch('https://automation.company.com/api/restart',
  {
    method: 'POST',
    headers: {
      'Authorization': `Bearer ${env.AUTOMATION_TOKEN}`,
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({
      host: hostId,
      service: serviceName,
      reason: `Dynatrace problem ${event.display_id}`
    })
  });

  return {
    success: response.ok,
    message: await response.text()
  };
}
```

#### Pattern 2: Clear Cache

```
tasks:
  - name: clear_cache
    type: dynatrace.http:request
    input:
      url: "https://{{ event().get('affected_entity_ids')[0]
    }}.internal/admin/cache/clear"
    method: POST
    headers:
      Authorization: "Bearer {{ env.ADMIN_TOKEN }}"
```

### Pattern 3: Traffic Redirect

```
// Redirect traffic away from failing instance
export default async function({ event }) {
  const failingInstance = event.root_cause_entity_id;

  // Update load balancer to remove failing instance
  await updateLoadBalancer({
    action: 'drain',
    instance: failingInstance,
    reason: event.display_id
  });

  return { redirected: true };
}
```

## 4. Kubernetes Remediation

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### Pod Restart

```
import { KubeConfig, CoreV1Api } from '@kubernetes/client-node';

export default async function({ event }) {
  // Extract pod info from problem
  const podName = extractPodName(event.root_cause_entity_id);
  const namespace = extractNamespace(event.root_cause_entity_id);

  // Configure kubernetes client
  const kc = new KubeConfig();
  kc.loadFromString(env.KUBECONFIG);
  const k8sApi = kc.makeApiClient(CoreV1Api);

  // Delete pod (Kubernetes will recreate it)
  try {
    await k8sApi.deleteNamespacedPod(podName, namespace);
    return {
      success: true,
      action: 'pod_restarted',
      pod: podName,
      namespace: namespace
    };
  } catch (error) {
    return {
      success: false,
      error: error.message
    };
  }
}
```



## Deployment Rollback

```
import { KubeConfig, AppsV1Api } from '@kubernetes/client-node';

export default async function({ event }) {
  const deploymentName = extractDeployment(event);
  const namespace = extractNamespace(event);

  const kc = new KubeConfig();
  kc.loadFromString(env.KUBECONFIG);
  const k8sApi = kc.makeApiClient(AppsV1Api);

  // Get deployment history
  const deployment = await k8sApi.readNamespacedDeployment(deploymentName,
    namespace);
  const currentRevision =
    deployment.body.metadata.annotations['deployment.kubernetes.io/revision'];

  // Rollback to previous revision
  await k8sApi.patchNamespacedDeployment(
    deploymentName,
    namespace,
    {
      spec: {
        rollbackTo: { revision: parseInt(currentRevision) - 1 }
      }
    },
    undefined,
    undefined,
    undefined,
    undefined,
    { headers: { 'Content-Type': 'application/strategic-merge-patch+json' } }
  );

  return {
    action: 'rollback',
    deployment: deploymentName,
    from_revision: currentRevision,
    to_revision: parseInt(currentRevision) - 1
  };
}
```

## Horizontal Pod Autoscaler Adjustment

```
export default async function({ event }) {
  // Temporarily increase replicas for high load
  const response = await fetch(
    `${env.K8S_API_SERVER}/apis/autoscaling/v2/namespaces/production/horizontalpo
dautoscalers/checkout-hpa`,
    {
      method: 'PATCH',
      headers: {
        'Authorization': `Bearer ${env.K8S_TOKEN}`,
        'Content-Type': 'application/merge-patch+json'
      },
      body: JSON.stringify({
        spec: {
          minReplicas: 5 // Increase from default 2
        }
      })
    }
  );

  return { scaled: response.ok };
}
```

## 5. Cloud Resource Remediation

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### AWS EC2 Instance Restart

```
import { EC2Client, RebootInstancesCommand } from '@aws-sdk/client-ec2';

export default async function({ event }) {
  const ec2Client = new EC2Client({
    region: 'us-east-1',
    credentials: {
      accessKeyId: env.AWS_ACCESS_KEY_ID,
      secretAccessKey: env.AWS_SECRET_ACCESS_KEY
    }
  });

  const instanceId = extractEC2InstanceId(event.root_cause_entity_id);

  const command = new RebootInstancesCommand({
    InstanceIds: [instanceId]
  });

  await ec2Client.send(command);

  return {
    action: 'ec2_reboot',
    instance_id: instanceId
  };
}
```

## AWS Lambda Function Redeploy

```
import { LambdaClient, UpdateFunctionCodeCommand } from '@aws-sdk/client-lambda';

export default async function({ event }) {
  const lambdaClient = new LambdaClient({ region: 'us-east-1' });
  const functionName = extractLambdaName(event);

  // Force cold start by updating function
  const command = new UpdateFunctionCodeCommand({
    FunctionName: functionName,
    S3Bucket: env.LAMBDA_BUCKET,
    S3Key: `${functionName}/latest.zip`
  });

  await lambdaClient.send(command);

  return { redeployed: functionName };
}
```

## Azure App Service Restart

```
export default async function({ event }) {
  const response = await fetch(
    `https://management.azure.com/subscriptions/${env.AZURE_SUBSCRIPTION_ID}/resourceGroups/${resourceGroup}/providers/Microsoft.Web/sites/${appName}/restart?api-version=2022-03-01`,
    {
      method: 'POST',
      headers: {
        'Authorization': `Bearer ${env.AZURE_TOKEN}`
      }
    }
  );

  return { restarted: response.ok };
}
```

## 6. Runbook Automation

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### Runbook Lookup and Execution

```
export default async function({ event }) {
  // Map problem types to runbooks
  const runbookMap = {
    'High CPU': 'runbook-cpu-investigation',
    'Memory exhaustion': 'runbook-memory-cleanup',
    'Disk space': 'runbook-disk-cleanup',
    'Connection pool': 'runbook-connection-reset'
  };

  // Find matching runbook
  let runbookId = null;
  for (const [pattern, id] of Object.entries(runbookMap)) {
    if (event.title.includes(pattern)) {
      runbookId = id;
      break;
    }
  }

  if (!runbookId) {
    return {
      action: 'no_runbook',
      message: 'No automated runbook for this problem type'
    };
  }

  // Execute runbook via automation system
  const result = await executeRunbook(runbookId, {
    problem_id: event.display_id,
    entity_id: event.root_cause_entity_id
  });

  return {
    action: 'runbook_executed',
    runbook: runbookId,
    result: result
  };
}
```

## Include Runbook Link in Notification

```
message: |
  :warning: *{{ event()['title'] }}*

  *Suggested Runbook:*
  {% if "CPU" in event()["title"] %}
  &lt;https://wiki.company.com/runbooks/cpu-investigation|CPU Investigation
  Runbook&gt;
  {% elif "Memory" in event()["title"] %}
  &lt;https://wiki.company.com/runbooks/memory-troubleshooting|Memory
  Troubleshooting Runbook&gt;
  {% else %}
  &lt;https://wiki.company.com/runbooks|Browse Runbooks&gt;
  {% endif %}
```

## 7. Approval Workflows

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### Human-in-the-Loop Pattern

```
tasks:
  # 1. Notify team of proposed remediation
  - name: request_approval
    type: dynatrace.slack:message
    input:
      channel: "#approvals"
      blocks:
        - type: section
          text:
            type: mrkdwn
            text: "*Remediation Approval Required*\n\nProblem: {{ event()
['title'] }}\nProposed Action: Restart pod\nEntity: {{ event()
['root_cause_entity_id'] }}"
          - type: actions
            elements:
              - type: button
                text:
                  type: plain_text
                  text: "Approve"
                  action_id: "approve_remediation"
                  style: primary
              - type: button
                text:
                  type: plain_text
                  text: "Reject"
                  action_id: "reject_remediation"
                  style: danger

  # 2. Wait for approval (timeout 30 min)
  - name: wait_approval
    type: dynatrace.automations:wait-for-event
    dependsOn: [request_approval]
    input:
      eventType: "remediation.approval"
      timeout: "30m"
      correlationId: "{{ event()['display_id'] }}"

  # 3. Execute if approved
  - name: execute_remediation
    type: dynatrace.automations:run-javascript
    dependsOn: [wait_approval]
```

```
conditions:
  - '{{ result("wait_approval").approved }}'
input:
  script: |
    // Perform remediation action
```

## Auto-Approve for Non-Production

```
conditions:
  - name: is_production
    expression: '{{ "Production" in event().get("management_zones", []) }}'

tasks:
  # Auto-execute for non-prod
  - name: auto_remediate
    conditions: [not is_production]
    # ... remediation action

  # Require approval for prod
  - name: request_prod_approval
    conditions: [is_production]
    # ... approval workflow
```

## Monitor Remediation Workflows

```
// Remediation workflow success rates
fetch events, from: now() - 7d
| filter event.type == "automation.workflow.execution"
| filter contains(workflow.name, "remediation") or contains(workflow.name, "auto-heal")
| summarize
  total = count(),
  succeeded = countIf(execution.status == "SUCCEEDED"),
  failed = countIf(execution.status == "FAILED"),
  by:{workflow.name}
| fieldsAdd success_rate = round(100.0 * succeeded / total, decimals: 2)
| sort total desc
```



```
// Problems auto-remediated vs manually resolved
fetch events, from: now() - 30d
| filter event.kind == "DAVIS_PROBLEM" and status == "CLOSED"
| summarize
    total = count(),
    auto_remediated = countIf(isNotNull(auto_remediation_id)),
    by:{time_bucket = bin(timestamp, 1d)}
| fieldsAdd auto_rate = round(100.0 * auto_remediated / total, decimals: 2)
| sort time_bucket asc
```

## Next Steps

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With remediation patterns in place, learn advanced integrations:

### Recommended Path

1. **WFLOW-08: JavaScript & HTTP Actions** - Custom integrations
2. **WFLOW-09: Security & Governance** - Production best practices

### Key Takeaways

- **Guardrails** prevent runaway automation
- **Rate limiting** avoids remediation loops
- **Kubernetes** can be automated via API
- **Runbook mapping** ties problems to solutions
- **Approval workflows** for production changes

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## Summary

In this notebook, you learned:

- Auto-remediation principles and maturity model
- Safety guardrails (rate limiting, time windows)
- Common remediation patterns
- Kubernetes pod/deployment remediation

- Cloud resource remediation (AWS, Azure)
  - Runbook automation integration
  - Human-in-the-loop approval workflows
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## References

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- [Workflow Actions](#)
  - [JavaScript Action](#)
  - [HTTP Request Action](#)
  - [Kubernetes API](#)
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